



S3. Learning Rate

Learning Rate determines how big a step the model takes toward learning.

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Among all the hyperparameters in Machine Learning and Deep Learning, there is one parameter that appears almost everywhere: **Learning Rate**.

This term is so important that you will encounter it not only in Deep Learning, but also in Machine Learning, Optimization, Numerical Methods, and many other computational fields.

To understand the idea, imagine that you are trying to walk toward a destination.

You start at your current location. Then you take a step. Then another. And another. The size of each step determines how quickly you reach your destination.

- If your steps are too small, reaching the destination takes forever.
- If your steps are too large, you might overshoot the destination and keep missing it repeatedly.

But here is an important detail that makes the analogy even more realistic.

- When people walk toward a destination, they rarely use the same step size the entire time. If the destination is far away, we naturally take larger steps to cover distance quickly.
- As we get closer, we tend to slow down and take smaller, more careful steps to avoid overshooting the target.

Optimization algorithms behave in a very similar way.

At the beginning of training, the model may be far from the optimal solution, so larger updates can help it learn quickly. As the model approaches the minimum loss, smaller updates become desirable because they allow the model to fine-tune its parameters and settle near the optimum rather than constantly jumping past it.

This idea is one of the reasons why Learning Rate is so important. It controls how large each update should be and, in many modern optimization methods, the effective step size can gradually adapt as training progresses.

Learning Rate works in exactly the same way. During training, a model gradually moves toward a better solution. The Learning Rate controls how large each movement should be.

If algorithms have **Time Complexity** to describe how efficiently they use computational resources, then optimization algorithms have **Learning Rate** to describe how aggressively they learn.

1. What is a Learning Rate?

A Learning Rate is a numerical value that determines how much the model updates its parameters during training. Every time the model makes a

prediction, it calculates an error.

Using that error, the optimizer decides how the weights should change. The Learning Rate determines the magnitude of that change.

For this reason, Learning Rate is sometimes called: Step Size; Update Size; Optimization Rate

Suppose we are standing 100 meters away from our destination. If we take:

- 1-meter steps, we need approximately 100 steps.
- 10-meter steps, we need approximately 10 steps.
- 50-meter steps, we might overshoot repeatedly.

The same idea applies to neural networks. A Learning Rate controls how quickly the model moves toward the minimum loss.

Mathematically, the weight update can be written as:

$$w_{new} = w_{old} - \eta \nabla L$$

Where:

- w represents the model weights
- ∇L represents the gradient
- η represents the Learning Rate

The Learning Rate acts as a multiplier for every update. A larger value creates larger movements. A smaller value creates smaller movements.

2. Learning Rate in Deep Learning

In Deep Learning, training occurs through a sequence of Epochs and Batches. Recall that:

- A Batch is a small subset of training data.
- An Epoch represents one complete pass through the training dataset.

During every batch:

1. Forward Propagation computes predictions.
2. Loss is calculated.

3. Backpropagation computes gradients.

4. The optimizer updates the weights.

And every single update uses the Learning Rate.

For example:

```
from tensorflow.keras.optimizers import Adam  
optimizer = Adam(learning_rate=0.001)
```

Here: `learning_rate=0.001` controls how aggressively the neural network updates its weights after every batch.

Notice something important. The Learning Rate is not applied once per epoch. It is applied every time the optimizer updates the model.

Therefore:

- More batches → more updates
- More updates → more Learning Rate applications

This is why Learning Rate and Batch Size often influence each other during training.

2.1. Learning Rate Too Small

Imagine using: `learning_rate = 0.000001`

The model takes extremely tiny steps. Training becomes very slow.

Although the model may eventually converge, it could require thousands of epochs to reach a good solution.

Typical symptoms:

- Slow learning
- Loss decreases very gradually
- Long training times

2.2. Learning Rate Too Large

Now imagine: `learning_rate = 1.0`

The model takes huge jumps. Instead of moving toward the minimum loss, it repeatedly overshoots it. Typical symptoms:

- Unstable training
- Oscillating loss curves
- Exploding loss values
- Failure to converge

In extreme cases, the loss may become larger and larger until training completely breaks down.

2.3. A Good Learning Rate

A good Learning Rate creates a balance. The model learns quickly while maintaining stable convergence.

Typical characteristics include:

- Smooth loss reduction
- Fast convergence
- Stable validation performance
- Consistent improvements across epochs

This balance is one of the most important goals in Deep Learning optimization.

3. Finding the Optimal Learning Rate

One of the biggest challenges in Deep Learning is determining the best Learning Rate. Unfortunately, there is no universal answer.

A Learning Rate that works perfectly for one problem may completely fail for another. For this reason, practitioners often experiment with multiple approaches.

3.1. Grid Search

The simplest approach is Grid Search. We define several candidate Learning Rates and test each one. For example:

```
learning_rates = [0.1, 0.01, 0.001, 0.0001]
```

Each Learning Rate is evaluated separately. The best-performing model is selected.

- Advantages: Simple and easy to understand
- Disadvantages: Computationally expensive and requires multiple training runs

3.2. Fixed Learning Rate

The most common beginner approach is to choose one Learning Rate and keep it constant throughout training. For example:

```
optimizer = Adam(learning_rate=0.001)
```

The value never changes.

- Advantages: Simple implementation and easy to explain
- Disadvantages: Not always optimal and may learn too aggressively later in training

3.3. Adaptive Learning Rate

Modern optimizers often adjust the Learning Rate automatically. Popular examples include: AdaGrad; RMSProp; Adam; AdamW

These optimizers monitor gradients and dynamically adjust update sizes. For example:

```
from tensorflow.keras.optimizers import Adam
```

```
optimizer = Adam()
```

Internally, Adam modifies the effective Learning Rate during training.

Advantages: Faster convergence; Less manual tuning; Works well in many practical applications

This is one reason why Adam has become one of the most widely used optimizers in Deep Learning.

3.4. Hyperparameter Optimization

For large projects, Learning Rate becomes part of a broader Hyperparameter Optimization process. Instead of manually testing values, automated systems search for good configurations.

Common approaches include:

- Grid Search
- Random Search
- Bayesian Optimization
- Evolutionary Algorithms

The goal is simple: ***Find the Learning Rate that produces the best validation performance while maintaining stable training.***

In modern Deep Learning projects, this process can save hundreds of hours of experimentation.

Summary

This section covers one of the most influential hyperparameters in Deep Learning: the Learning Rate. We learned that Learning Rate determines how large each optimization step should be during training. A value that is too small causes extremely slow learning, while a value that is too large may prevent the model from converging altogether.

We also explored how Learning Rate interacts with epochs and batches, why modern optimizers such as Adam use adaptive strategies, and how techniques

such as Grid Search and Hyperparameter Optimization help us find effective values.

Understanding Learning Rate is essential because even the most powerful neural network can fail if its step size is poorly chosen.

References

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